

Data-driven Economics
MSc in Data Science
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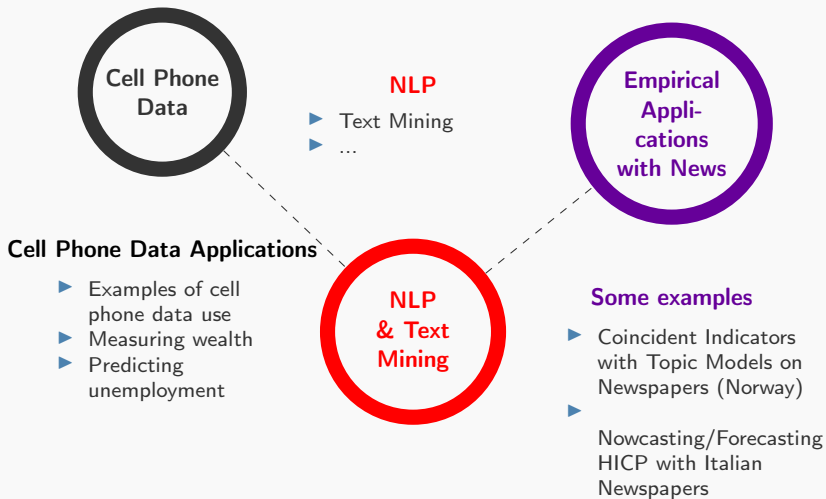
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University of Rome - La Sapienza
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What we have seen last time

1. Bodas et al. Credit/Debit card transactions and retail sales
2. Ardizzi, Emiliozzi, Marcucci and Monteforte, News and Consumer Card Payments $\text{EPU} \Rightarrow \text{Payments} \Rightarrow \text{Consumption}$
3. Payments , Card Transactions and COVID-19 lockdown effects on consumption
4. $\text{D'Amuri \& Marcucci (2017)}$: Google Trends for nowcasting & forecasting

This lecture in one slide



Data-driven Economics: Using cell phone data

- We saw that a typical feature of **data-driven economics** and **Big data** is the possibility of **using granular/unstructured Big data to measure economic phenomena** or **to nowcast/predict economic or financial variables**.
- In the last lecture we saw some applications of **debit/credit card transaction data**
- Now we are going to see a couple of empirical applications which use **cell phone data**

Cell Phone Data Applications in Macroeconomics

Estimating Economic Characteristic with Phone Data

By Blumenstock (AER-PP, 2018)

Blumenstock (AER-PP, 2018)

“Estimating Economic Characteristic with Phone Data”

- Blumenstock (2018) tries to ask the question of **whether the “digital footprints” of an individual can be used to infer his or her socioeconomic characteristics**
- In particular Blumenstock (2018) tries to **predict the wealth of an individual using her/his mobile phone records** to **then aggregate** these phone-based predictions **into accurate national statistics** generalizing the results of Blumenstock (2014) and Blumenstock, Cadamuro, and On (2015)
- <https://doi.org/10.1257/pandp.20181033>

Blumenstock (AER-PP, 2018)

“Estimating Economic Characteristic with Phone Data”

- Blumenstock (2018) adopts **two samples** from **two countries**:
- **856 respondents** to a phone survey conducted in **Rwanda in 2009**
- 20-minute phone survey with a geographically stratified sample in 2009
- **1,234 respondents** to a face-to-face survey conducted in **Afghanistan in 2015-2016**
- Face-to-face and phone-based interviews (two provinces, Kabul and Parwan)
- The author tries to use the “digital footprints” left through the phone use (text messages, phone calls, social media, etc.)
- Goal: estimate individual's i wealth Y_i from data X_i that summarizes i 's history of mobile phone use.

Blumenstock (AER-PP, 2018)

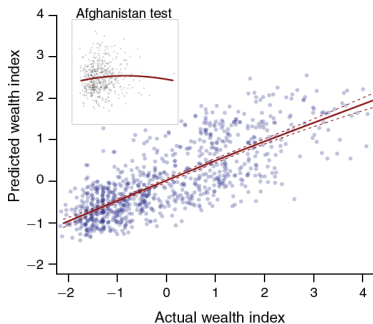
“Estimating Economic Characteristic with Phone Data”

- **2 steps:**
 - **1st step:** transform digital device data into a vector of K metrics $X_i = (x_{i1}, \dots, x_{iK})$ which quantifies call duration, number of unique cell towers used by i , etc.
 - **2nd step:** fit a model $Y_i = f(X_i)$ that captures the relationship between the target variable and the vector of phone use metrics.
- Fitting $f()$ using a flexible supervised algorithm like **the Gradient Boosting**
- **Only two months of mobile phone activity** immediately prior to the date of the survey is used to construct X_i

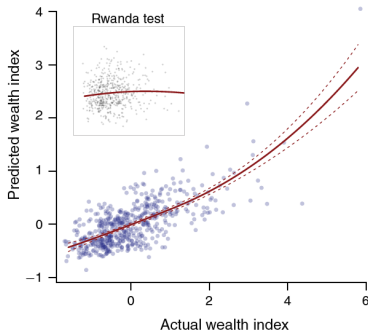
Blumenstock (AER-PP, 2018)

“Estimating Economic Characteristic with Phone Data”

Panel A. Rwanda model



Panel B. Afghan model



$$R^2 = 0.33$$

Blumenstock (AER-PP, 2018)

“Estimating Economic Characteristic with Phone Data”

- The **model predicts very well** individual wealth (average $R^2 = 0.33$)
- But **a model trained on one country does not work in the other one** (e.g. applying the model trained on Rwandan survey to predict Afghan survey respondents do almost as well as random guessing, $R^2 = 0.07$, and 0,05)
- **Great potential!**
- By 2020, roughly **three quarters of the world's population** (5.7 billion people) **will own a mobile phone**
- Many applications can be created to estimate economic characteristics from mobile phone use at a fraction of cost for a traditional country-wide survey (Census).
- For example, **measurement!** Or for **consumer profiling, targeting, market segmentation**. It can be used to **develop credit scores** (Francis, Blumenstock, and Robinsong, 2017; Bjorkegren and Grissen, 2015)

Blumenstock (AER-PP, 2018)

“Estimating Economic Characteristic with Phone Data”

- Message to take home:
- Blumenstock (2018) managed to **approximate the wealth of people in Rwanda and Afghanistan using just two months of such call detail records** (CDR, e.g. records of phone calls, text messages, airtime purchases, mobile money use, location of cellphone tower nearest to both parties, and other user data).
- Blumenstock (2018) achieved the **same accuracy of a five-year-old national census** at a **tiny fraction of the cost**.
- Main idea: a person's wealth is correlated with mobile transactions
- His model can approximate wealth level of **30%** of entire population in Afghanistan and **10%** of the population in Rwanda.

Cell Phone Data Applications in Macroeconomics

Tracking Employment Shocks Using Mobile Phone Data

By Toole, Lin, Muehlegger, Shoag, Gonzalez and Lazer (Journal of the Royal Social Interface, 2015)

Toole et al. (2015) “Tracking Employment Shocks Using Mobile Phone Data”

- Toole et al. (2015) show that **data from mobile phones can be used to measure and observe economic shocks.**
- They present a **novel method to detect massive layoffs**, identifying individuals affected by them and **predicting changes in aggregate unemployment rates using call details records (CDR) from mobile phones**
- Jameson L. Toole, Yu-Ru Lin, Erich Muehlegger, Daniel Shoag, Marta C. González and David Lazer (2015), “Tracking employment shocks using mobile phone data”, Journal of the Royal Society Interface
- <http://dx.doi.org/10.1098/rsif.2015.0185>

Toole et al. (2015) “Tracking Employment Shocks Using Mobile Phone Data”

- Toole et al. (2015) observe **changes in calling behavior** following a **big plant's closure**
- They observe **significant declines in social behavior** and mobility **following job loss**
- Using these features identified **at the micro level**, they show that the **same changes** in these calling behaviors, once **aggregated** at the regional level, **can improve forecasts of macro unemployment rates**

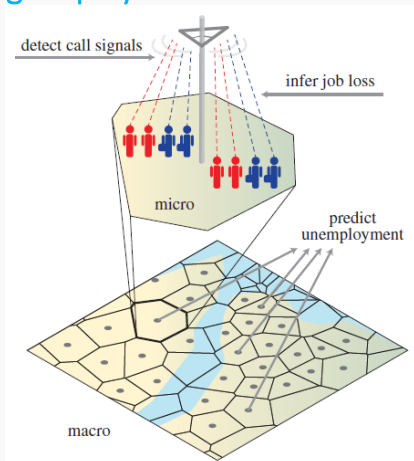
Toole et al. (2015) “Tracking Employment Shocks Using Mobile Phone Data”

- **Current measurement draw on limited sources** of information.
- **New methodology** to infer changes in the macroeconomy in **near real time, at arbitrarily fine spatial granularity**, using data passively collected from mobile phones.
- Data from **two European countries**
- In the **first country**, a **massive layoff** from **one big plant closure** allow to identify mobile phone behavior for people laid off (**mobility** and **social interactions**)

Toole et al. (2015) “Tracking Employment Shocks Using Mobile Phone Data”

This is the **idea**:

- Use calling behavior of individuals to infer job loss
- Measure these variables and include them in prediction of unemployment at the macroeconomic level



Toole et al. (2015) “Tracking Employment Shocks Using Mobile Phone Data”

- In the **second country**, **macro-level data are available** and the authors show that **changes in mobility and social behavior predict unemployment rates** ahead of official reports and more accurately than traditional forecasts.
- **DATA:** at the community level **15-month period between 2006 and 2007** in an undisclosed European country with **15%** market share and **with a large-scale layoff event**
- At the province level CDR data from **another European country** with approximately **20%** market share and **36 months of data from 2006 to 2009**
- Each dataset include an anonymous ID for caller and callee, the location of the tower through which the call was made, and the time the call occurred.

Toole et al. (2015) “Tracking Employment Shocks Using Mobile Phone Data”

- The **large-scale layoff** occurred in December 2006 (closure of an auto-parts manufacturing plant) in a city with **three mobile phone towers** within close proximity of the town and the plant.
- Roughly **1,100 workers** were laid off in a **small town of 15,000**.
- Assign weights through Bayesian model based on changes in mobile phone activity.
- Compute the **conditional probability that a user is a non-resident worker laid off**.

Toole et al. (2015) “Tracking Employment Shocks Using Mobile Phone Data”

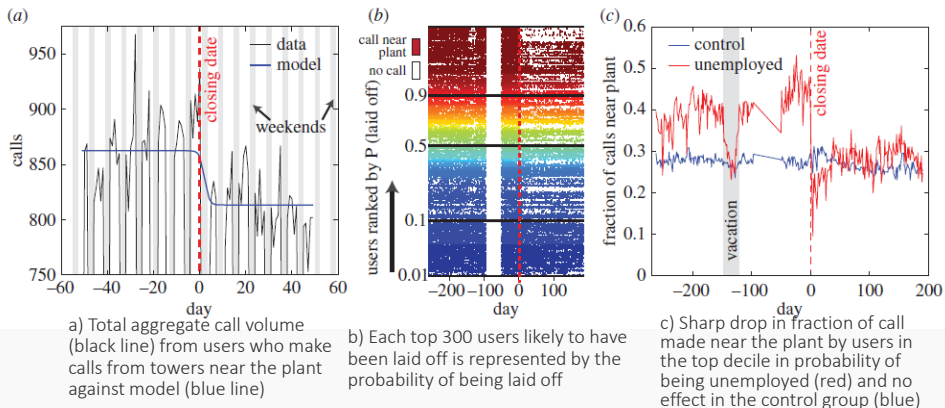
- **6 monthly variables for social behavior:**

1. Total calls
2. Number of incoming calls
3. Number of outgoing calls
4. Calls made to individuals physically located in the town of the plant
5. Number of unique contacts
6. Fraction of contacts called in the previous month not called in current one (referred as churn)

- Also **3 measures of changes in mobility:**

1. Number of unique locations visited
2. Radius of gyration
3. Average distance from most visited tower

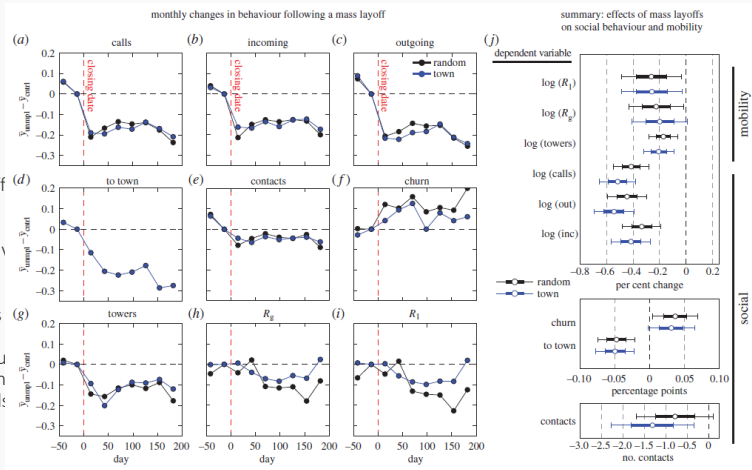
Toole et al. (2015) "Tracking Employment Shocks Using Mobile Phone Data"



Toole et al. (2015)

Changes in social networks and mobility following layoffs

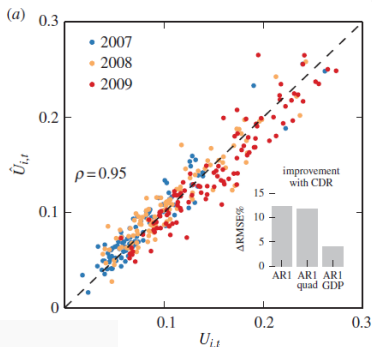
- Calls made by laid off people drop 51%
- Asymmetric drop (outgoing calls -54%, -41% incoming)
- # of unique contacts called after laid off is significantly lower
- Month-to-month churn +3.6 percentage point
- Social interaction falls after laid off



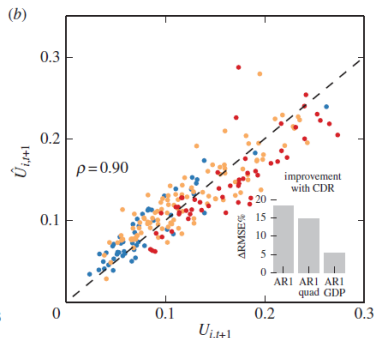
Toole et al. (2015)

Predicting Unemployment Rates

- Compare predictions with AR1, AR1 Quad (with quadratic term), AR1 GDP (AR1 with GDP as leading indicator)
- Adding metrics from mobile phone data improves forecast accuracy, reduces RMSE of UR between 5 and 20%
- Using mobile phone data from only the first half of the quarter (before surveys are conducted) achieve 3-10% improvement



a) Nowcasting the unemployment rate for the same quarter (t). Strong correlation (0.95)



b) Forecasting the unemployment rate for the next quarter (t+1). Strong correlation (0.85)

NLP & Text Mining

NLP & Text Mining

By Various Authors

What is NLP?

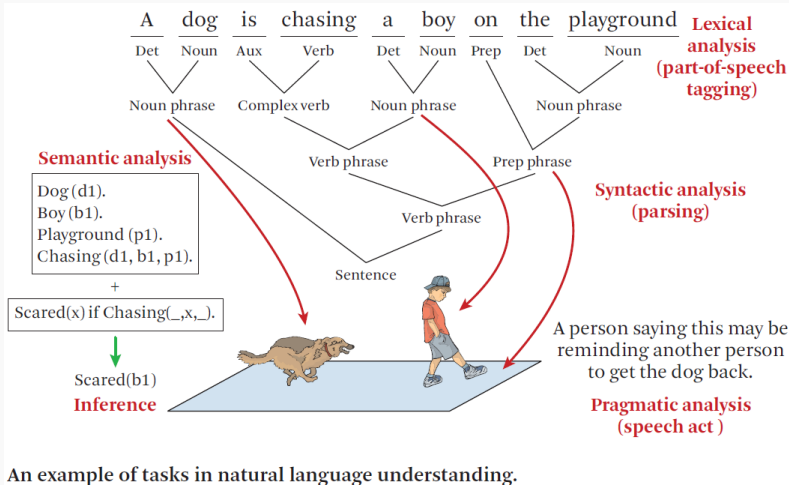
- **NLP** is concerned with **developing computational techniques to enable a computer to understand the meaning of natural language text.**
- Humans can easily understand the context of text data because natural language is designed to make human communication efficient.
- In contrast, a programming language is designed to facilitate a computer understanding
- NLP is difficult because:
 1. A lot of **“common sense” knowledge** is omitted
 2. There are a lot of **ambiguities** and the reader/hearer is assumed to know how to resolve them.

What is NLP?

- To **enable a computer to make sense of text** some tasks are needed:
 1. **Lexical analysis:** to understand the meaning of basic units of a language (words). E.g. can use word boundaries for English but what about Chinese?
 2. **Syntactic analysis:** to understand how words are related with each other in a sentence
 3. **Semantic analysis:** to determine the meaning of a sentence
 4. **Pragmatic analysis:** to determine the meaning in a context
 5. **Discourse analysis:** to analyze a large chunk of text

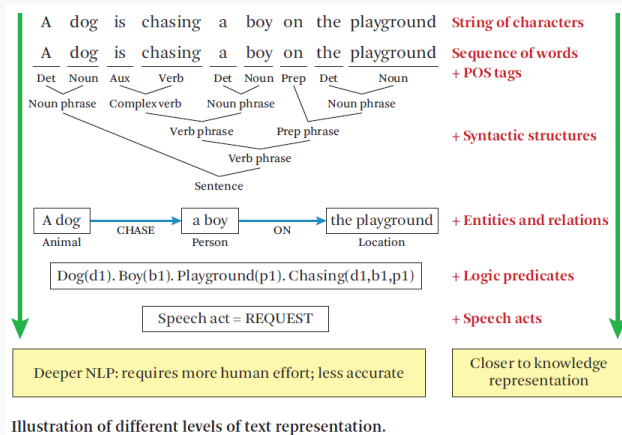
What is NLP?

Tasks



From Zhai and
Massung (2016)

What is NLP? Text Representation



From Zhai and Massung (2016)

What is NLP?

1. Question answering
2. Machine Translation
3. Sentiment Analysis
4. Natural Language Generation: Summarization
5. Natural Language Generation: Conversation/Dialogue
6. Natural Language Generation: Image Captioning
7. Natural Language Generation: Video Captioning
8. Natural Language Generation: Visual Question Answering
9. Automatic Speech Recognition

Language Modeling and Generation

Language Modeling

- A **statistical language model (LM)** is a **probability distribution over word sequences**

$$P(\mathbf{w}) = P(w_1, \dots, w_n)$$

- A LM gives any sequence of words a potentially different probability.
- The purpose is to usually assign high weights to plausible sentences, e.g., in speech recognition or machine translation
- Also used for language generation now (predict next word given previous words), especially with new Recurrent Neural Networks (RNN) models

Traditional N-gram LMs

N-gram

- Use chain rule to generate words left-to-right

$$P(w_1, \dots, w_n) = \prod_i P(w_i | w_1, \dots, w_{i-1})$$

- You can't condition on the **entire left context** ($w_1, \dots, w_{i-1}$)

$P(??? | \text{Turn to page 113 and look at the picture of the})$

- N-gram models make a **Markov assumption**

$$P(w_1, \dots, w_n) = \prod_i P(w_i | w_{i-k}, \dots, w_{i-1})$$

E.g. :

$P(\text{please close the door}) = P(\text{please} | \text{START})P(\text{close} | \text{please}) \dots P(\text{STOP} | \text{door})$

Traditional N-gram LMs

- How do we know $P(w|history)$?
 - Use **statistics** from data (examples using **Google N-Grams**)
 - E.g. what is $P(door | the)$?

Training Counts

198015222 the first
194623024 the same
168504105 the following
158562063 the world

14112454 the door

23135851162 the *

$$\hat{P}(\text{door}|\text{the}) = \frac{14112454}{23135851162}$$
$$= 0.0006$$

- This is the **maximum likelihood** estimate

Sparsity Issue & Parameter Estimation

- We get **new words** all the time (e.g. *antiestablishmentarianism*, *kakorrhaphioophobia*, *wwwxyzabc156.com*,...)...and worse for new bigrams and trigrams!

➤ **Maximum likelihood** estimate will not get us very far

$$\hat{P}(w_i|w_{-1}) = \frac{c(w_{-1}|w)}{\sum_{w'} c(w_{-1}|w')}$$

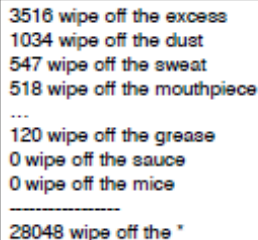
➤ Need to **smooth** these estimates

➤ General method (procedurally)

- Take your empirical counts
- Modify them in various ways to improve estimates

➤ General method (mathematically)

- Often can give estimators a formal statistical interpretation
- ...but not always
- Approaches that are mathematically obvious aren't always what works



3516 wipe off the excess
1034 wipe off the dust
547 wipe off the sweat
518 wipe off the mouthpiece
...
120 wipe off the grease
0 wipe off the sauce
0 wipe off the mice

28048 wipe off the *

Smoothing Techniques

- We often want to make **estimates from sparse statistics**:

- $P(w \mid I \text{ denied the})$

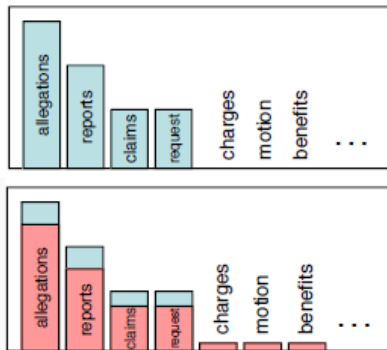
- 3 allegations
 - 2 reports
 - 1 claims
 - 1 request
 - 7 total

- **Smoothing** flattens **spiky distributions** so they generalize better:

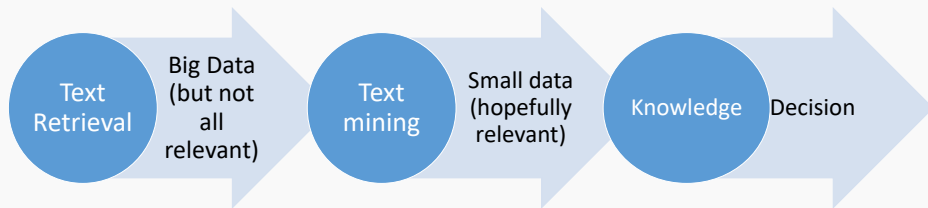
- $P(w \mid I \text{ denied the})$

- 2.5 allegations
 - 1.5 reports
 - 0.5 claims
 - 0.5 request
 - 2 other
 - 7 total

- **Very important all over NLP**, but easy to do badly!
Be careful!



Text Retrieval and Text Mining



Text data is special

- **Text data** is generally **produced by humans**, **rather than by computers or sensors**
 - In contrast with e.g. image data
- **Text data** is generally meant to be **consumed by humans**, **rather than by computers or sensors**
 - So-called unstructured data

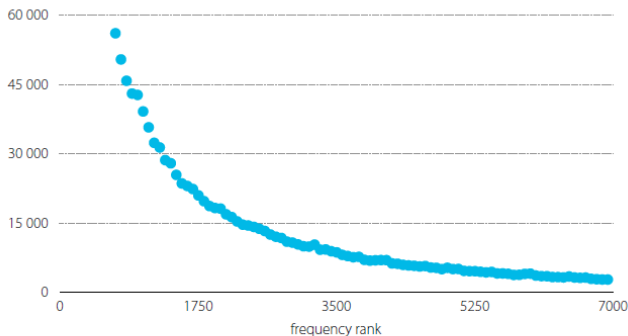
Zipf's Law

In textual data, the frequency of a given word is inversely proportional to its frequency rank.

The r^{th} most frequent word has a frequency $f(r)$ that scales according to $f(r) \propto \frac{1}{r^\alpha}$, $\alpha \approx 1$

Zipf(1936, 1949)

Zipf's law

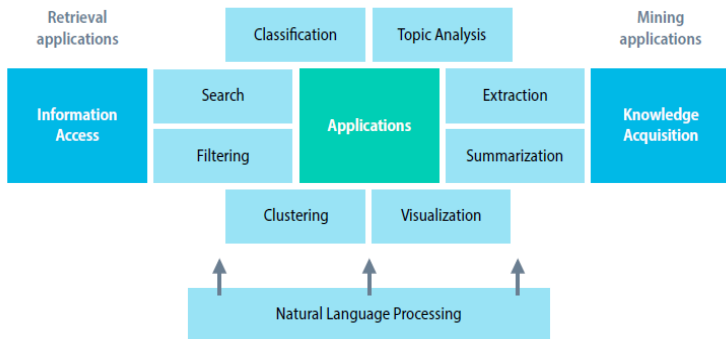


Word frequency data from the [Contemporary American English Corpus](#)

Typical applications of text mining

- **Search:** Take a user's query and return relevant documents.
- **Filtering:** Filter a stream of incoming documents.
- **Classification:** Sort documents into predefined categories.
- **Clustering:** Discover groups of similar text documents.
- **Topic Analysis:** Identify topics in a document collection.
- **Visualization:** Visually display patterns in text data.
- **Information Extraction:** Extract entities and relations between them.
- **Summarization:** Generate a summary of a document collection.

Text mining



Adapted from Zhai and Massung (2016)

Two Functions

➤ Information Access

- Enable the user to access relevant information in time
 - *Search engines (pull), recommender systems (push)*

➤ Knowledge Acquisition

- Enable the user to acquire knowledge “hidden” in text
 - *Information extraction, topic analysis*

Two Perspectives

➤ **Natural Language Processing**

- Make limited inferences based on the natural language text
 - *information extraction*

➤ **Data Mining**

- Discover and extract interesting patterns in the text data
 - *topic modelling*

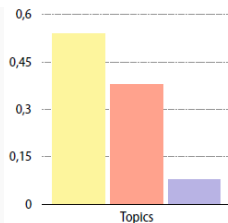
Topic Models

How many genes does an organism need to survive? Last week at the genome meeting here, two genome researchers with radically different approaches presented complementary views of the basic genes needed for life. One research team, using computer analyses to compare known genomes, concluded that today's organisms can be sustained with just 250 genes, and that the earliest life forms required a mere 128 genes.

human
genome
dna
genetic
genes
sequence
gene
molecular
sequencing
map
information
genetics
mapping
project
sequences

evolution
evolutionary
species
organisms
life
origin
biology
groups
phylogenetic
living
diversity
group
new
two
common

computer
models
information
data
computers
system
network
systems
model
parallel
methods
networks
software
new
simulations



Source: Blei (2012)

Three Stages for NLP

- **Retrieving from textual data**
 - Information Retrieval
- **Analysing the linguistic structure**
 - Natural Language Processing
- **Building statistical models**
 - Statistical Modeling

Text data is special

- Text data is generally produced by humans, rather than by computers or sensors
 - In contrast with e.g. image data
- Text data is generally meant to be consumed by humans, rather than by computers or sensors
 - So-called unstructured data

Thorsrud (JBES, 2018)

Words are the New Numbers: A Newsy Coincident Index of the Business Cycle

By Leif Thorsrud

Journal of Business & Economic Statistics, 2018

What is the state of the economy today?

- ▶ The answer is important because...
 - ▶ Policy makers need to know the state of the economy in real time to devise appropriate policy responses
 - ▶ Forecasters would benefit from conditioning on a timely information set
- ▶ ...but, it is a difficult question
 - ▶ In real time, our best known measure of economic activity, GDP growth, is not observed:
 - ▶ It is registered on a quarterly frequency
 - ▶ It is published with a considerable lag

Traditional approaches

Use:

- ▶ National Account statistics and high-frequency (financial) data from structural databases
- ▶ Mixed-frequency time-series models (and model combination)

Result in:

- ▶ Coincident indexes: Stock and Watson [1988], Mariano and Murasawa [2003], Aruoba et al. [2009], Marcellino et al. [2013]
- ▶ Nowcasting: Giannone et al. [2008], Breitung and Schumacher [2008], Angelini et al. [2011], Kuzin et al. [2011]

Problems and opportunities

Problems:

- ▶ Few high-frequency variables reflecting the broader economy
- ▶ Instabilities across time (Stock and Watson [2003])

Opportunities:

- ▶ New technologies allow for alternative models and data sources, like (newspaper) text

Text as data?

1. Potentially available at a high frequency
2. Potentially reflecting the broader economy
3. A number is a fact, but the media in which it is presented/discussed/opinionated adds to the information
4. Rational (in)attention theory: Media an important channel for information diffusion

But, putting text into models is somewhat new (to economists) - until recently...

We show

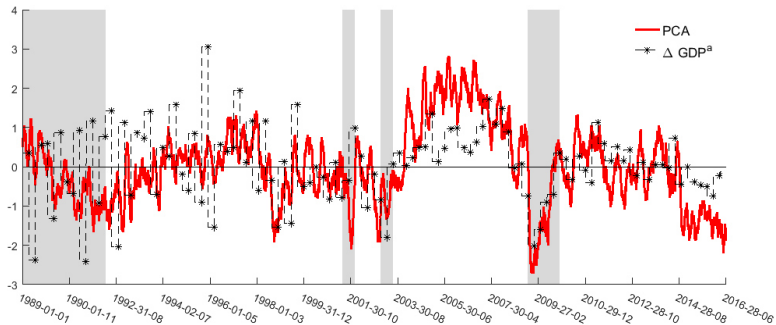
When textual news is appropriately classified using a topic model - it adds valuable information for understanding economic fluctuations

In particular, it can be used to:

1. Track the business cycle very well
2. Nowcast GDP growth

A short (in-sample) preview

Simple PCA decomposition of topics (unsupervised) and GDP growth



Key input:



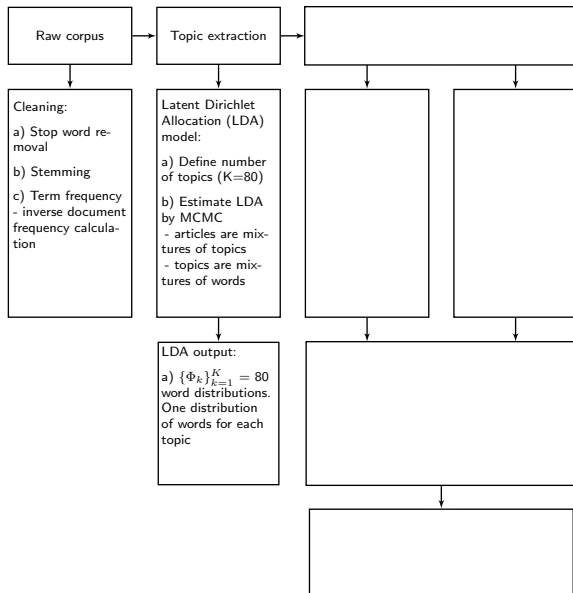
Use a primary source for new information directly - the newspaper!

Key hypothesis:

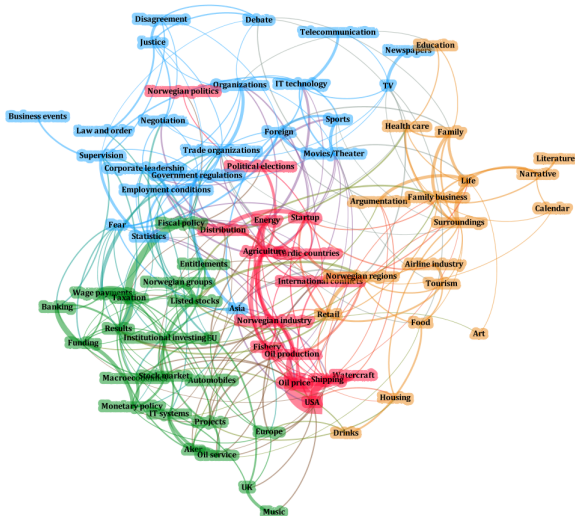
The more a newspaper writes about a topic the more likely it is that this topic reflects something of importance for the economy's current and future needs and developments

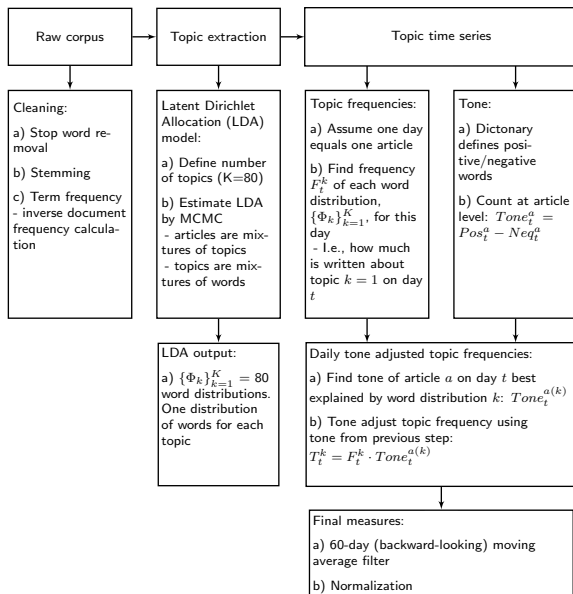
Two-step approach

1. Decompose the newspaper corpus into tone adjusted topic frequency time series
 - ▶ Cleaning the corpus
 - ▶ Finding the topics
 - ▶ Constructing time series
2. Develop a time-varying mixed frequency factor model to:
 - ▶ Estimate a daily news-based coincident index
 - ▶ Nowcast



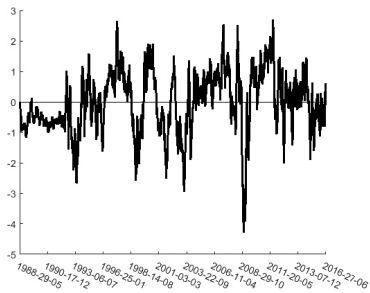
A topic representation of the whole corpus



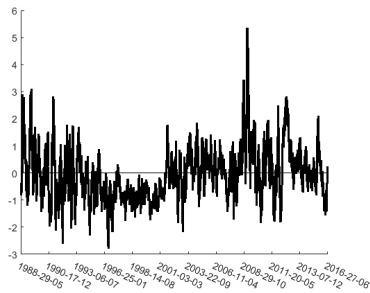


Topic time series

Topic: Monetary Policy (72)



Topic: Funding (42)



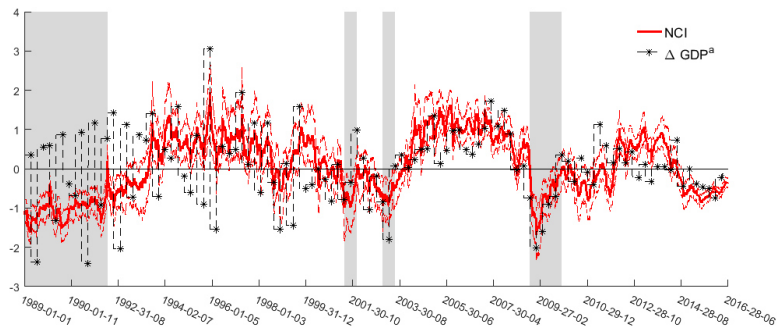
The Dynamic Factor Model

- ▶ y_t = quarterly GDP growth and daily newspaper topics
- ▶ Time aggregation using cumulator variable(s) (Harvey [1990] and Banbura et al. [2013])
- ▶ Estimation: Bayesian MCMC

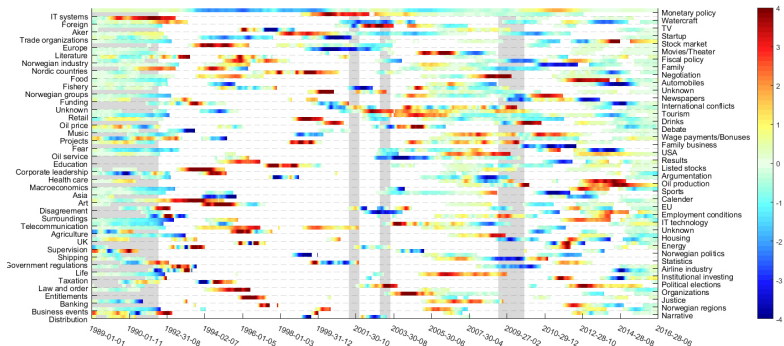
Two extensions

1. Time-varying parameters and threshold dynamics enforcing dynamic sparsity (Nakajima and West [2013])
 - ▶ Why: News topics might be informative about macro developments in some periods, but perhaps not always
2. Mixture Auxiliary Particle Filter
 - ▶ Why: For fast and efficient updating (nowcasting)

A Newsy Coincident Index (NCI)



News and index decompositions



Large degree of time-varying sparsity.
Topic contributions “tell” a plausible narrative

Summary 1

In-sample:

1. Almost perfect classification properties
2. News performs better than traditional economic indicators/models
3. Combined usage of news and time-varying parameters adds value
4. Provides high-frequency information about what drives/reflects economic fluctuations

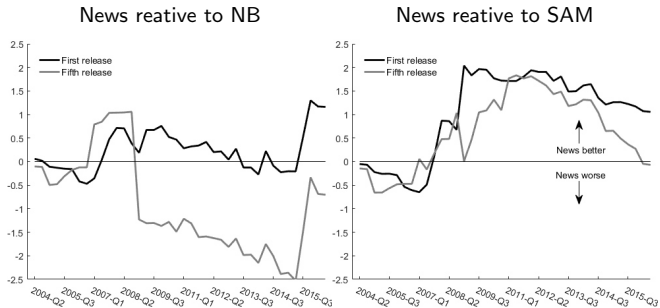
Nowcasting evaluation (out-of-sample)

RMSFE (using real-time data): Fifth release

Type	Nowcast for T_q				Backcast for T_q		
	$T_q - 2$				$T_q - 1$		
Quarterly information							
Daily information	$T - 80$	$T - 60$	$T - 40$	$T - 20$	T	$T + 10$	$T + 30$
RMSFE	0.537	0.538	0.547	0.520	0.497	0.502	0.500
Cum. imp. (%)		0.304	1.958	-3.153	-7.304	-6.472	-6.801
Relative RMSFE							
NCI^{notvp} / NCI	1.287**	1.324**	1.220	1.243**	1.107	1.105	1.104
NCI^{nosw} / NCI	1.083**	1.076***	1.092**	1.106**	1.074***	1.074***	1.077***
CI / NCI	1.113	1.106	1.072	1.241**	1.080	1.076	1.102
AR / NCI	1.142	1.139	1.121	1.491***	1.557**	1.543***	1.549***
RW / NCI	1.696***	1.692***	1.665***	1.553***	1.622***	1.608***	1.614***
SAM / NCI	1.020	0.974	0.934	1.008	1.000	0.991	0.996
NB / NCI					0.961	0.953	0.958

Nowcasting performance better than, or equally good as,
alternatives

News versus NB and SAM: Cumulative squared prediction error differences



Good around turning points?

Summary 2

Out-of-sample:

1. News superior to simple benchmarks, and equally good as sophisticated benchmarks and judgment
2. Combined usage of news and time-varying parameters adds value
3. Policy makers and the statistical office would have benefited from using news

Conclusion

1. The usage of newspaper data is novel and improves classification power and nowcasting
2. Enforcing sparsity and allowing for time-varying parameters in the DFM is important
3. News reduces noise

Part of a larger research agenda using text as data and news.
Similar news topics as used here:

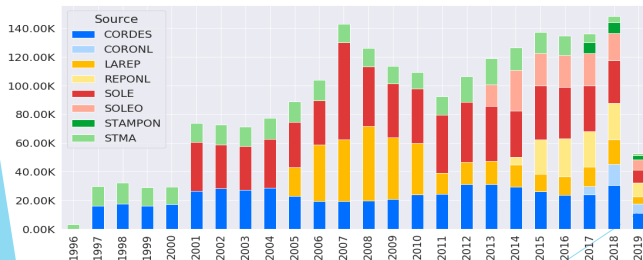
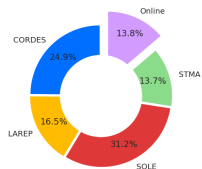
- ▶ Predict cross section of daily stock returns, explain continuation patterns, and yield significant risk adjusted returns (Larsen and Thorsrud [2017])
- ▶ Explain quarterly long-run productivity developments (Larsen and Thorsrud [2015])

Forecasting inflation with news

Deriving indicators from a large corpus of Italian documents

*By M. Bernardini, P. Cariello, M. De Leonardis, J. Marcucci, F. Quarta, A. Tagliabracci
Bank of Italy, mimeo 2019*

Query



Deriving Indicators from a large corpus of Italian documents

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Text Mining Pipeline

We worked only on 'Inflation' subquery, removing online sources (~300k docs)

Cleaning the corpus

- Lower case
- Remove numbers
- Remove punctuation
- Tokenization
- Filter
- Dictionary for LDA

Finding the topics

- LDA (Latent Dirichlet Allocation)
- Select best model

Constructing time series

- Daily topic intensity
- Sentiment Analysis
- Finding correlations
- Time series forecasting

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Cleaning's phase



	Raw text	Unique words	Identify collocations	Remove stopwords	Stemming	TF-IDF adjustment	Min DF
Number of words	221,080,503	508,605	508,952	492,246	294,104	123,315	9,942

- Lower case
- Remove numbers
- Remove punctuation
- Tokenization (split words)
- Stopwords (Italian & English)
- Remove common names and surnames (from 'byline' field)
- Remove words with length < 3 or > 26
- Lemmatization/Stemming
- Bigram/Trigram
- TF-IDF or ZIPF's Law

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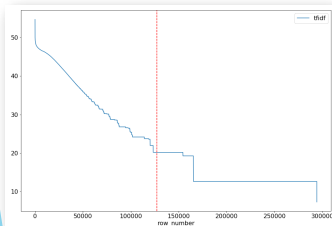


Building the Dictionary

TF-IDF

$$w_{i,j} = tf_{i,j} \times \log \left(\frac{N}{df_i} \right)$$

TF/IDF (25% totale; DF>300) → ~10.000 words

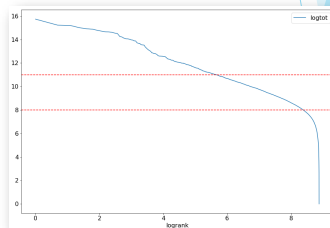


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ZIPF's LAW

"the frequency of any word is inversely proportional to its rank in the frequency table"

Zipf Law ($8 < \log(\text{freq}(w)) < 11$) → ~3.500 words



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Latent Dirichlet Analysis

- Finding the LDA Model with best K (Number of Topics)
- List of K values used [10, 20, 30, 50, 80]
- TrainSet [80%] - TestSet [20%]
- Optimizer: Online variational Bayes
- Alpha (Doc-Concentration) = uniformly $(1.0 / K)$ [default]
- Beta (Topic-Concentration) = $(1.0 / K)$ [default]
- Evaluation Metrics:
 - Topic Perplexity (*how model captures the distribution of the held out set*)
 - Topic Coherence (*the degree of semantic similarity between its high scoring words*)



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Topic models

- ▶ All topic models are based on the same basic assumptions:
 - ▶ each document consists of a mixture of topics
 - ▶ each topic consists of a collection of words
- ▶ In other words, topic models are built around the idea that the semantics of our document are actually being governed by some hidden, or “latent” variables that we are not observing.
- ▶ As a results, the goal of topic modeling is to uncover these latent variables – **topics** – that shape the meaning of our document and corpus.

Latent Dirichlet Allocation (LDA)

Some Intuition

- ▶ Imagine you have three topics:

Topic 1 $\Rightarrow$ politics

Topic 2 $\Rightarrow$ fashion

Topic 3 $\Rightarrow$ sports

- ▶ Assume you have a recipe, i.e. 50% of Topic 1, 30% of Topic 2 and 20% of Topic 3
- ▶ Each topic consists of a set of words times their probabilities $word_j \times \Pr_{j,k}$
- ▶ You take this recipe and generate a document based on this model rules
- ▶ You obtain a document with words from each of the three topics with different frequencies.
- ▶ But, in reality it is the reverse!
- ▶ You have to learn a topic model from a set of words appearing in a set of documents

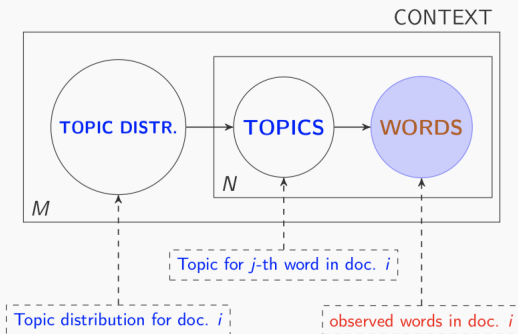
Latent Dirichlet Allocation (LDA)

Some Intuition

- ▶ Important Choices:
 - ▶ **How many topics?**
 - ▶ How are the topics assigned to a document?
- ▶ LDA is a BOW (Bag-of-Word) model so there is no syntax rule!
- ▶ LDA Model uses a real document using all words assuming they are related and tries to learn a model that would explain how such document would have been generated from the beginning if we followed that scenario

Latent Dirichlet Allocation (LDA)

- N words for M documents



Latent Dirichlet Allocation (LDA)

Some Intuition

- ▶ Extra rules:
 - ▶ **Hyperparameter α**
 - ▶ **Hyperparameter β**
- ▶ α controls the **per document-topic distribution**
- ▶ β controls the **per topic-word distribution**
- ▶ A **high α** means that every document is likely to contain a mixture of most of the topics and not just any single topic
- ▶ A **low α** means that a document is more likely to be represented by just a few of the topics.
- ▶ A **high β** means that each topic is likely to contain any mixture of most of the words and not just any single words specifically.
- ▶ A **low β** means that a topic may contain a mixture of just a few of the words.

Latent Dirichlet Allocation (LDA)

Some Intuition

- ▶ In a nutshell:
- ▶ A **high** α will make document to appear more similar to each others.
- ▶ A **high** β will make the topics to appear more similar to each others.
- ▶ The final result:
- ▶ We get a file that contains all the topics made of all the words with their probabilities to belong to that topic.
- ▶ How many topics?
- ▶ It depends on the application domain.

Latent Dirichlet Allocation (LDA)

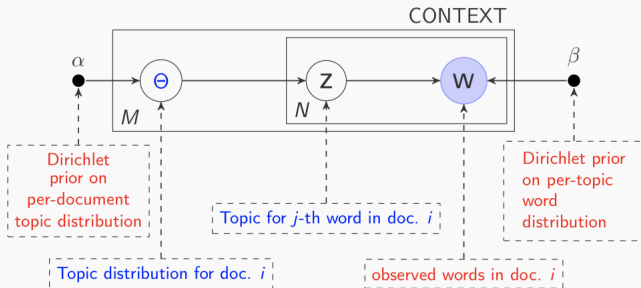
Some Intuition

- ▶ Think of LDA as a little robot guy head which has an understanding of words, topics and context and combines them all together.
- ▶ A document is a probability distribution over topics
- ▶ A topic is a probability distribution over words
- ▶ LDA model takes text and reads the BOW
- ▶ Remember that the text corpus and pre-processing are the more critical steps.

Latent Dirichlet Allocation (LDA)

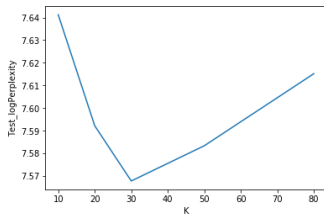
- N words for M documents

$$P(W, Z, \theta, \varphi, \alpha, \beta) = \prod_{i=1}^K P(\varphi_i; \beta) \cdot \prod_{j=1}^M P(\theta_j; \alpha) \cdot \prod_{t=1}^N P(z_{j,t} | \theta_j) \cdot P(w_{j,t} | \varphi, z_{j,t})$$



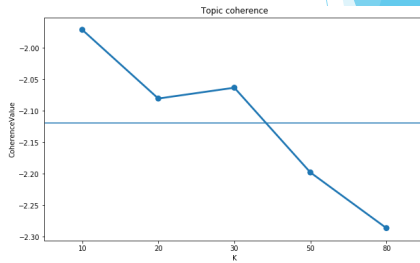
LDA metrics

Perplexity



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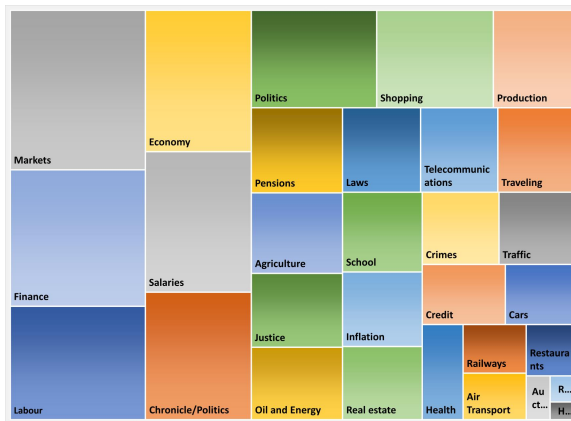
Coherence



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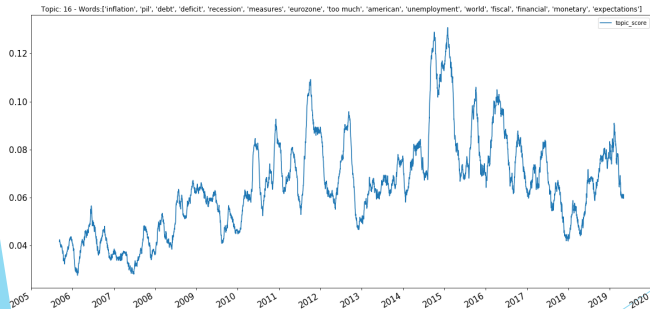
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Topics concentration



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Topic 16 - Intensity



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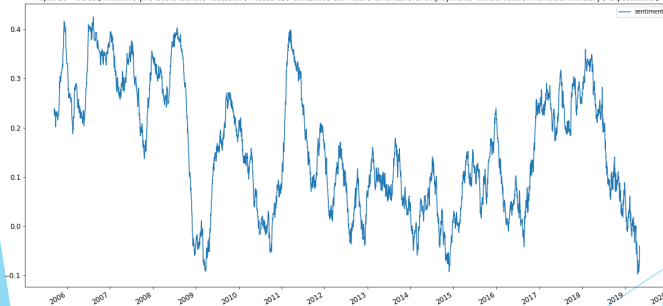
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Topic 16 - Sentiment



Topic: 16 - Words ['inflation', 'pił', 'debt', 'deficit', 'recession', 'measures', 'eurozone', 'too much', 'american', 'unemployment', 'world', 'fiscal', 'financial', 'monetary', 'expectations']

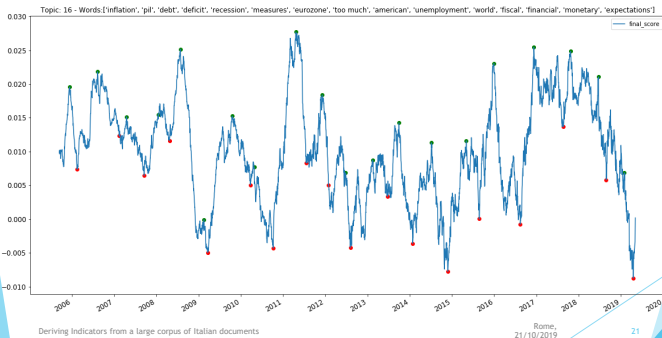


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Topic 16 - Intensity * Sentiment

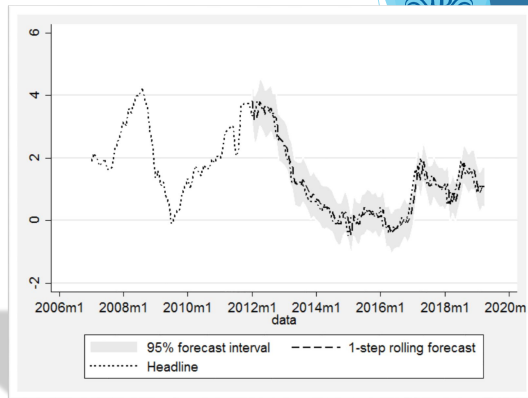


Forecasts

HICP (Harmonized Index of Consumer Prices) = Year-on-year rate of change at the monthly frequency.

Out-of-sample exercise on a rolling window of five years.

Forecast for Headline Inflation (HICP) and the relative prediction obtained with a AR(1)-X model (Benchmark).



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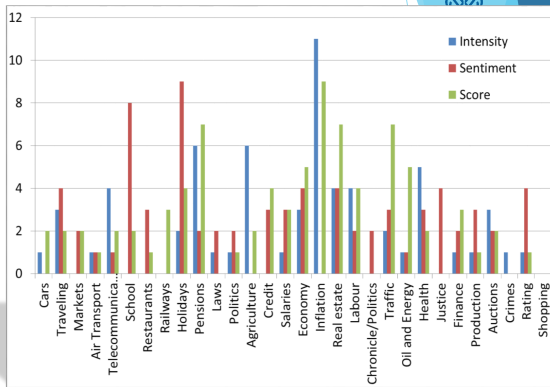
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Predictive Power



Y-Axis: number of times each news topic on the x axis helps predicting the inflation rate with respect to an AR(1) benchmark using a rolling window of 60 months.

We depict a different bin for each different topic measure (intensity, sentiment and score, which is the intensity weighted for the score).



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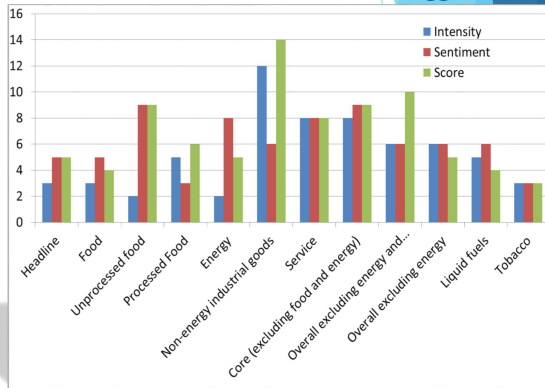
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Predictive Power

Y-Axis: how many topic-based models outperform the benchmark using intensity, sentiment or the score of each topic.

As highlighted in the previous literature for other languages - see for example (Thorsrud, 2018) - weighting the intensity indicators using sentiment or tonality does indeed help in predicting the variable of interest.



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What we have seen

1. 2 empirical applications where authors use cell phone data to measure macro variable such as wealth in real time or to nowcast/forecast unemployment
 - ▶ Blumenstock (2018) - Cell phone data and wealth in Afghanistan and Rwanda
 - ▶ Toole et al. (2015) - Cell phone data and nowcasting/forecasting unemployment in 2 European countries
2. NLP & Text Mining introduction
3. 2 applications of text mining in economics
 - ▶ Thorsurd (2018) - Text and Norwegian business cycle (Newsy Coincident Indicator)
 - ▶ Bernardini et al. (2019) - Text and Nowcasting HICP Inflation